MARY LETEY

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EDUCATION

Sept 2023 — Present	**Harvard University**	PhD — **Applied Mathematics**
Sept 2022 — June 2023	**Perimeter Institute**	MSc — **Theoretical Physics**
Sept 2018 — June 2022	**University of Cambridge**	BA Honours — **Mathematics**

PUBLICATIONS & MANUSCRIPTS

∗ *denotes equal first authorship;* † *denotes equal senior authorship.*

In-context classification: Calibration and noise in linear attention with softmax readout
PriGM Workshop at Neurips 2026 • Rysmakhanov*, **M. Letey***, Lu, Zavatone-Veth, Pehlevan

What can linear attention learn in-context from nonlinear teachers?
PriGM Workshop at Neurips 2026 • **M. Letey***, Rysmakhanov*, Lu, Pehlevan, Zavatone-Veth

Sequential correlations change in-context learning: Effective context length and architectural mismatch
HiLD Workshop at ICML 2026 • **M. Letey**, Lu, Pehlevan†, Zavatone-Veth†

Pretrain–test task alignment governs generalization in in-context learning
ICLR 2026 • **M. Letey**, Zavatone-Veth, Lu, Pehlevan

Theory of scaling laws for in-context regression: Depth, width, context and time
ICLR 2026 • Bordelon, **M. Letey**, Pehlevan

Asymptotic theory of in-context learning by linear attention
Proceedings of the National Academy of Sciences (2025) • Lu, **M. Letey**, Zavatone-Veth, Maiti, Pehlevan

Quantum initial conditions for curved inflating universes
Physical Review D (2024) • **M. Letey***, Shumaylov*, Agocs, Handley, Hobson, Lasenby

Discrete symmetry in Lorentzian spaces
Master's thesis, Perimeter Institute (2023) • Advisor: Latham Boyle

WORK & RESEARCH EXPERIENCE

Research Internship Winter 2026	**Hudson River Trading** • New York, USA Machine learning for market prediction
Research Internship Summer 2023	**Quebec Artificial Intelligence Institute** • Montreal, Canada Equivariant architectures for neural PDE solvers
Research Internship Summer 2022	**Cavendish Laboratory** • Cambridge, UK Research on early-universe cosmology and curved spacetimes
Internship Summer 2020	**Embryo Ventures** • London, UK Due diligence and founder outreach
Research Internship Fall 2019 — Spring 2020	**Iguana Investments** • Cambridge, UK Market prediction models

INVITED TALKS

Johns Hopkins — Data Science Seminar, October 2026 — Invited talk on finite-data effects in in-context learning

EPFL — FLARE Workshop, May 2026 — Invited talk on theoretical models of in-context learning

SIAM — Conference on Uncertainty Quantification, March 2026 — Invited talk: "Asymptotic theory of in-context learning"

Kempner Institute — Spring Into Science Retreat, May 2024 — Opening talk: "Asymptotic theory of in-context learning"

TEACHING & ADVISING

26-27 **Research mentor** • Matthew Chin — KURE-funded undergraduate research project

2026 **Research mentor** • Arman Rysmakhanov — Summer mentorship resulting in two papers

25-26 **Preconcentration advisor** • Applied Mathematics, Harvard University

2025 **Teaching Fellow** • APMATH 50, Harvard University

21-23 **Tutor** • Mathematics, physics, and university admissions, Blue Education

AWARDS & FUNDING

2025	**Graduate Fellowship**	Kempner Institute — Full support for graduate studies
2024	Clare Marie Doris Innovation Fund	Harvard University
2023	**Graduate Prize Fellowship**	Harvard University
	Summer Research Bursary	McGill University
2022	**International Scholar Award**	Perimeter Institute — Full support for MSc
	Summer Research Bursary	Royal Society